



Hand-Computer Control Interface

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Why gesture control interface?

- More natural interaction
- More immersive experience in AR/VR
- Lower learning curve & accessibility



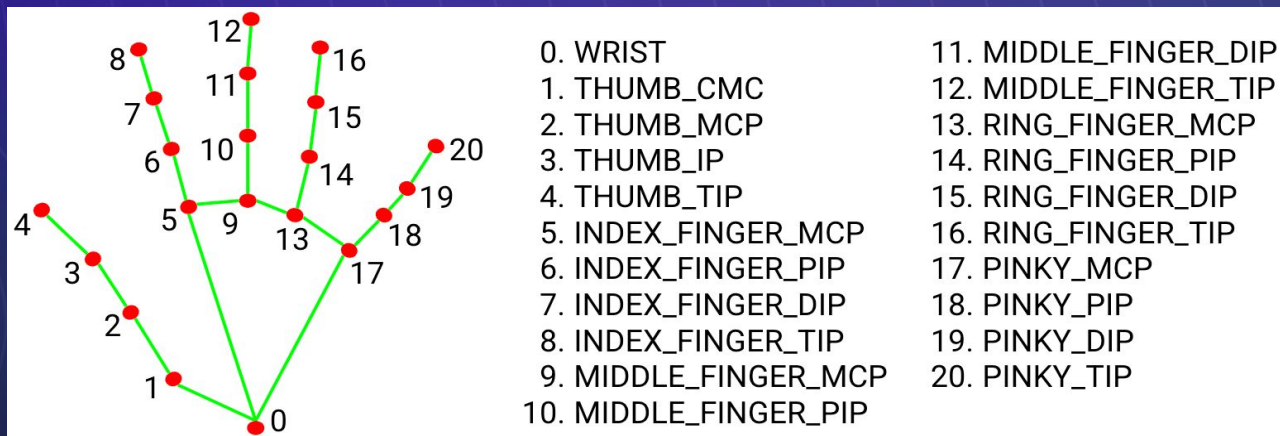
Computer Controls

- FIST: Neutral, stop all actions
- POINT: Left-click down
- PEACE: Right-click down
- THUMBS_UP: Scroll up
- THUMBS_DOWN: Scroll down
- FIVE: Double left-click
- Hand position: Move cursor
 - Absolute
 - Relative



Landmark Extraction

- Capture video frame by frame
- Extract landmarks with Mediapipe (Google CV framework)
- 21 (x, y) coords



Dataset Overview

- Custom dataset
 - Keystrokes to log which gesture (indexed 0 to 5)
 - Data saved to CSV file
 - Implicit data augmentation
-
- 180 data points
 - 30 data points per label

```
GESTURE_MAP = [  
    Gesture.FIST,          # ID 0  
    Gesture.POINT,         # ID 1  
    Gesture.PEACE,         # ID 2  
    Gesture.THUMBS_UP,     # ID 3  
    Gesture.THUMBS_DOWN,   # ID 4  
    Gesture.FIVE,          # ID 5  
]
```

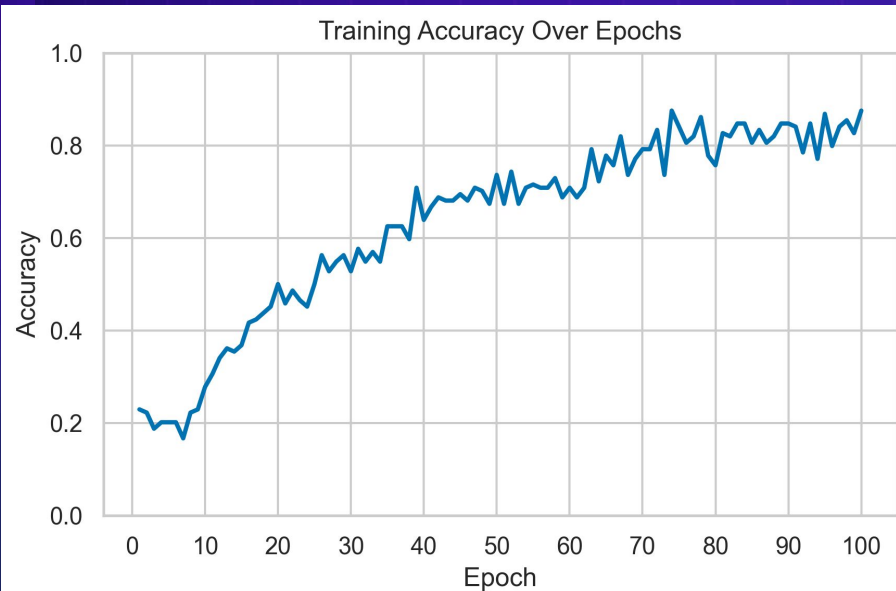
ML Model & Training

- MLP model
 - Compress 42 features to 6 gestures
 - Dropout regularization
 - ReLU activation
- Training
 - 100 epochs
 - Adam optimizer
 - Cross-entropy loss

```
class KeyPointModel(nn.Module):  
    def __init__(self, num_classes):  
        super().__init__()  
        self.net = nn.Sequential(  
            nn.Dropout(0.2),  
            nn.Linear(42, 20),  
            nn.ReLU(),  
            nn.Dropout(0.4),  
            nn.Linear(20, 10),  
            nn.ReLU(),  
            nn.Linear(10, num_classes)  
        )  
  
    def forward(self, x):  
        return self.net(x)
```

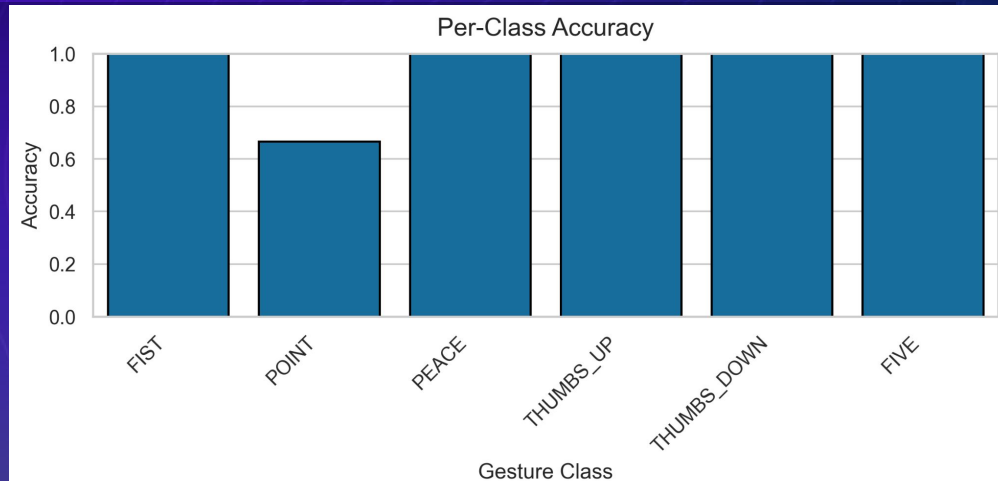
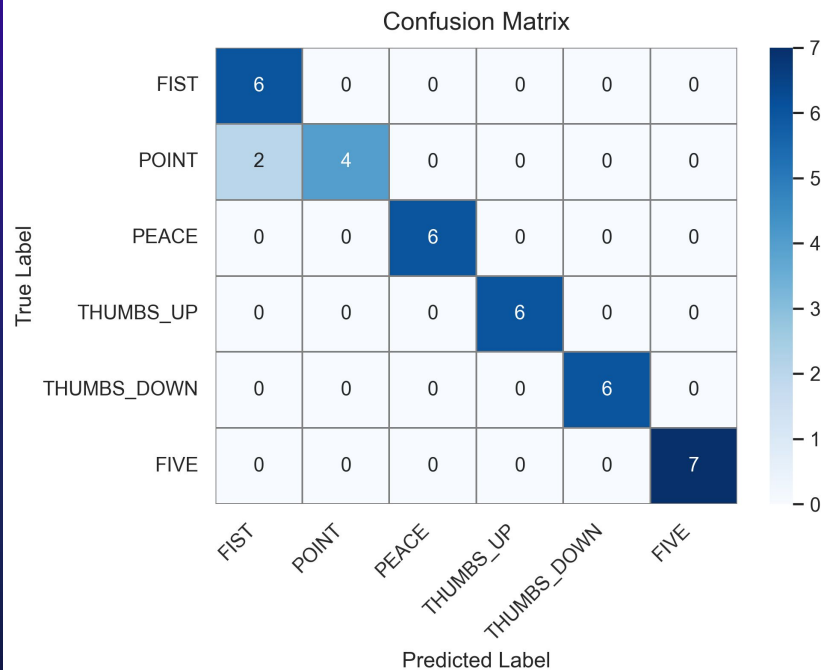
Results

Epoch 100/100 Loss: 0.4222 Acc: 0.8750



Test Accuracy: 0.9459

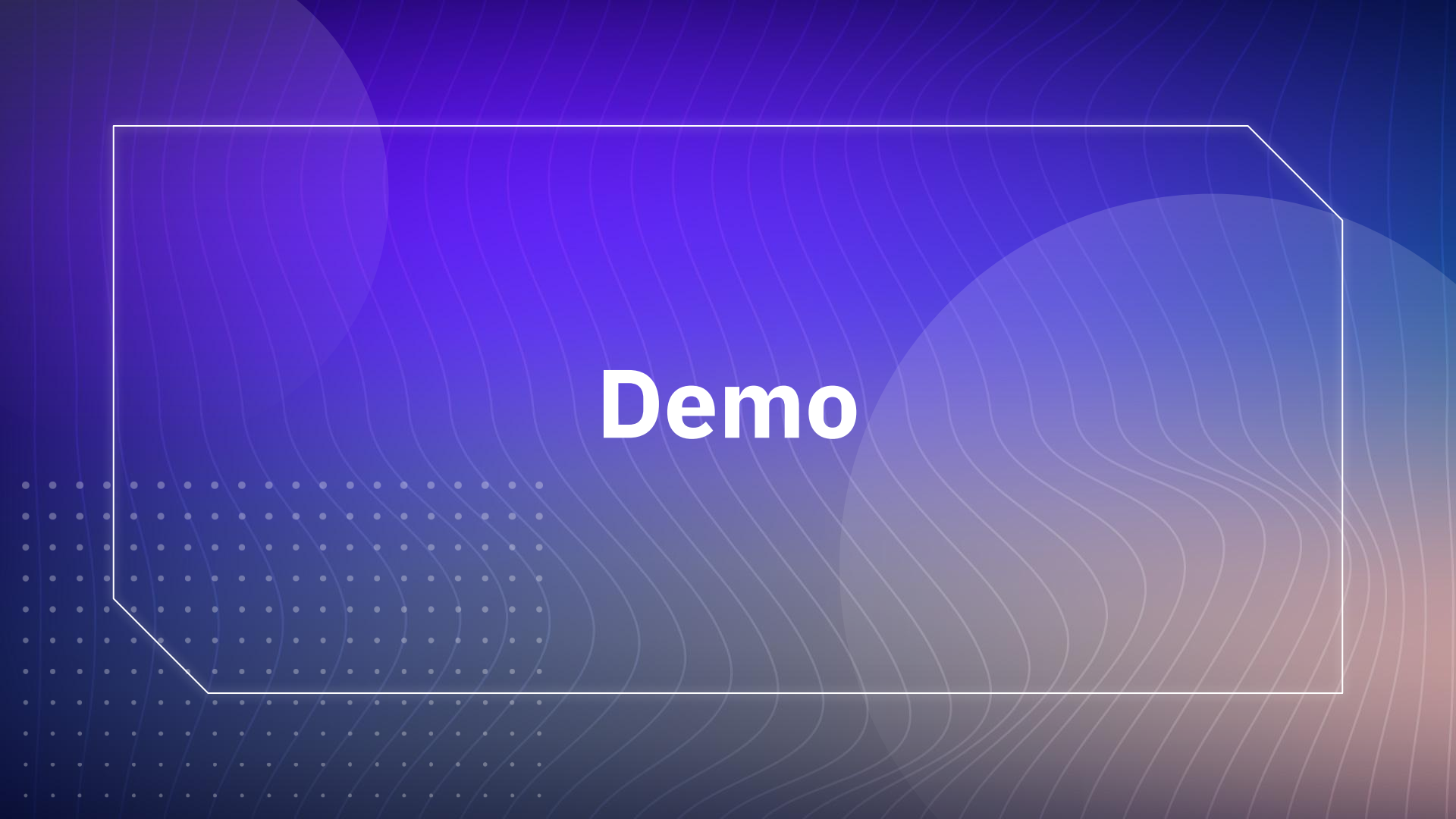
Results Cont.



```
FIST : 1.000
POINT : 0.667
PEACE : 1.000
THUMBS_UP : 1.000
THUMBS_DOWN : 1.000
FIVE : 1.000
```


Limitations

- Size of our dataset not as extensive as preset due to creating our own custom dataset
- Due to collecting data in a controlled environment, our model might have reduced accuracy in generalization
 - Lighting
 - Camera Angles
 - Background
 - Distance
 - Camera Quality
 - Hand Variability



Demo



Thank you!